Ahmed T. Hammad

PhD, Causal Machine Learning | AI & Decision Making | Program & Policy Evaluation
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PROFESSIONAL SUMMARY

PhD-level impact evaluation specialist and AI/ML researcher with 10+ years of experience in econometric and machine learning data modelling, designing, conducting, and supporting rigorous experimental and quasi-experimental evaluations across health, social policy, food security, and environmental domains. Deep expertise in causal machine learning, integration of AI and ML into evaluation workflows, and SDMx for official statistics data exchange. Author of six published open source tools and peer-reviewed publications in PNAS, PLOS One, and STOTEN.

CORE SKILLS

Impact Evaluation: RCT design, quasi-experimental methods, difference-in-differences, synthetic control, propensity score matching, instrumental variables, uplift modelling

Causal ML: Causal forests, double/debiased ML, X-Learner, CUPED, heterogeneous treatment effects, CATE estimation, DoWhy, EconML, CausalML

AI & ML: Supervised/unsupervised learning, deep learning, NLP, LLMs, RAG pipelines, agentic systems, sequential/online learning

Statistics: Econometrics, Bayesian methods, probabilistic modelling, copulas, bandit algorithms, time series

Programming: R (expert), Python (expert), Stata, SQL

Tools: scikit-learn, TensorFlow, PyTorch, H2O, MLflow, Darts, FastAPI, Docker, DuckDB, GCP, Streamlit

Domains: Social policy, public health, food security, environmental monitoring, education, climate change and adaptation

Languages: English (fluent), Arabic (native), Italian (native), Indonesian (basic)

PROFESSIONAL EXPERIENCE

Data Science Consultant | *Baffelan OÜ, Estonia* | 2026 – Present

- Contributing causal ML and AI expertise supporting open data portals, SDMx, and LLM governance frameworks for government clients.

Data Science Consultant | *Green Heart Care, Australia* | May 2025

- Led end-to-end design and delivery of a production AI/NLP system for a regulated NDIS healthcare provider, processing hundreds of caregiver notes daily with automated multi-class anomaly flagging for clinical review.
- Managed all decisions on model selection, pipeline architecture, and performance evaluation. Established data governance standards and ethical AI guidelines for sensitive patient data.

Lead Data Science Lecturer | *Le Wagon, Singapore & Bali* | Apr 2022 to Present

- Designed and delivered intensive applied ML and AI programmes to 200+ international students across Singapore, Bali, Australia, and Malaysia. Consistently ranked top 1-2 globally on student evaluations.
- Led twice a two-week enterprise data science intensive for 100+ professionals at SLB (Schlumberger) as part of a three-year, 500-person workforce transformation programme.
- Full curriculum spans Python, R, SQL, supervised and unsupervised learning, deep learning, NLP, causal inference, AI, and production ML deployment.

Guest Editor | *IOP Publishing, Environmental Research: Infrastructure & Sustainability* | Sep 2024 to Feb 2025

- Coordinated peer review and editorial direction for a focus issue on geospatial and data-driven methods for climate resilience and infrastructure impact assessment.

Data Science Consultant | *PT. Mantra Bali, Indonesia* | Apr 2020 to Apr 2021

- Built a computer vision system for real-time quantification of landfill waste volumes, designing the end-to-end ML pipeline from image capture through model inference and operational reporting.

Data Science Consultant | *Simile, Timor-Leste* | Feb 2020 to Jan 2021

- Developed predictive models for climate-induced early warning systems, building geospatial alert pipelines integrating satellite and sensor data for real-time flood risk signalling with NGO partners.

Data Science Consultant | *Tripper, Indonesia* | Jul 2019 to Aug 2019

- Designed crop-yield prediction models using remote sensing and time-series data to support agricultural pricing and planning for smallholder farmers.

Research & Fieldwork Manager | *Catholic University of the Sacred Heart & Ipsos, Milan* | 2013 to 2018

- Managed large-scale randomised controlled trial evaluations at national scale, including experiment design, treatment assignment, data collection, and causal effect estimation.
- Coordinated 20+ fieldwork supervisors and 10,000+ respondents across three multi-wave studies: Family St.A.R. and OpenTeQ. Delivered clean datasets and quantitative reports to academic and government clients.
- Contributed to Ipsos impact assessments of public communication campaigns, handling questionnaire design and quantitative reporting.

OPEN SOURCE TOOLS

- onlineCML (Python): streaming causal inference library implementing online estimators for IPW, Augmented IPW, Overlap Weighting, meta-learner frameworks, and a Causal Hoeffding Tree.
- Argenta (Python): warehouse-native causal ML experimentation tool integrating with Snowflake, BigQuery, and Redshift. Applies Causal Forest, X-Learner, CUPED, and sequential testing inside the data warehouse.
- syntCF (R): causal effect estimator combining synthetic control with ML for programme and policy evaluation in observational time series.
- pbox (R): probabilistic risk assessment tool for computing marginal, joint, and conditional probabilities across correlated variables.
- memoire (Python): persistent causal knowledge graph for AI systems, tracking directed causal relationships with confidence scores.
- pygcapi / rgcapi (Python/R): the only libraries interfacing with the Gain Capital API for automated trading execution.

CERTIFICATIONS

- An introduction to SDMx for data producers - SDMx vocabulary for beginners - SDMx vocabulary for data managers — SIS-CC .Stat Academy 2026
- Winter School in Social Network Analysis — University of Greenwich, London, 2018
- Winter School in Fundamentals and Methods for Impact Evaluation of Public Policies — FBK-IRVAPP, Trento, 2016

EDUCATION

PhD, Causal Machine Learning | *Catholic University of the Sacred Heart, Milan* | 2018 to 2022

Thesis: Causal Machine Learning for Program and Policy Evaluation. Focus on heterogeneous treatment effect estimation, synthetic counterfactual methods and ML augmentation of classic econometric estimators.

Visiting Researcher | *University of Leeds, UK* | Nov 2018 to Apr 2019

Developed a web-based analytical framework integrating socio-demographic data with electronic petition records.

MSc, Sociology and Social Research | *University of Trento, Italy* | 2015 to 2017

Thesis: Applied modern mediation analysis to estimate indirect causal effects of healthcare access using experimental data from the Oregon Health Insurance Experiment.

BSc, Sociology | *University of Milan-Bicocca, Italy* | 2010 to 2014

Thesis: Assessed effects of Family Group Conference programme on student well-being through randomised evaluation.

SELECTED PUBLICATIONS

Falchetta, G. & Hammad, A.T. (2025). Tracking green space along streets of world cities. Environmental Research: Infrastructure and Sustainability.

Shayegh, S. et al., incl. Hammad, A.T. (2023). Prioritizing COVID-19 vaccine allocation: an AI-enabled geospatial decision-support framework. PLOS One.

Hammad, A.T. & Falchetta, G. (2022). Probabilistic forecasting of cropland vegetation health. Science of the Total Environment.

Falchetta, G., Hammad, A.T. & Shayegh, S. (2020). Planning universal accessibility to public healthcare in sub-Saharan Africa. PNAS